

Where We Are: An Audit of the AR(2) Extension to the Space-time Skew- t Model of Morris et al. (2017)

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Abstract

We audit an AR(2) extension of the space-time skew- t model of [Morris, Reich, Thibaud, and Cooley \(2017\)](#). The extension generalises the AR(1) prior placed on three transformed tail parameters ($\mathbf{w}_{tk}^*$, z_{tk}^* , σ_{tk}^{2*}) to a stationary AR(2). After a careful re-reading of the paper, the production R code, and the Metropolis–Hastings update mechanics, we identify three deviations from the stationary AR(2) prior: (i) a $t = 2$ conditional density that is inconsistent with the simulator’s joint initial draw; (ii) a missing lag-2 factor in the Metropolis–Hastings ratio for each Y_t update; and (iii) an apparent asymmetry in the ϕ proposal. Detailed-balance analysis confirms (i) and (ii) as real bugs and refutes (iii): the original silent-rejection scheme for ϕ already targets the correct posterior, and the “correction” we initially applied biases the chain toward the boundary of the stationarity region. The final state of the audit is that (i) and (ii) have been patched and (iii) has been reverted, with five regression tests covering the structural and behavioural assertions. We also point to an optional follow-up: refactor the ϕ update to mirror the truncated-proposal scheme used in the AR(1) reference code if mixing near the stationarity boundary becomes a concern.

1 Introduction

[Morris, Reich, Thibaud, and Cooley \(2017\)](#) introduce a space-time skew- t process for threshold exceedances. Section 3.2 of that paper extends the spatial-only model to the spatiotemporal setting by placing AR(1) priors on three transformed parameters $\mathbf{w}_{tk}^*$, z_{tk}^* , σ_{tk}^{2*} , while preserving the marginal skew- t distribution at every time point. A natural extension replaces the AR(1) prior with a stationary AR(2). The implementation lives in `code/R/ar2/` and was constructed by analogy to the AR(1) reference code in `code/R/ar2/update_params.R`.

This note has three goals:

1. State the AR(2) prior used in the implementation and derive its induced moments and conditional factorisation (Section 2).
2. Identify the three deviations between the production code and the stationary AR(2) prior, and apply a careful detailed-balance analysis to each (Section 3).
3. Describe the patches applied, the tests that guard them, and the outstanding work (Section 4–Section 5).

The central result is that two of the three deviations are real bugs and have been patched: the missing lag-2 term in the Metropolis–Hastings ratio of Y_t (Issue 2) and the inconsistent $t = 2$ conditional density that broke the simulator-versus-MCMC agreement on the joint of (Y_1, Y_2) (Issue 1). The third (Issue 3) is not a bug at all – the original silent-rejection scheme is a textbook-valid sampler, and the correction we initially applied to it introduces a target-shifting bias of its own, so that correction has been reverted.

2 Model setup

2.1 Notation and the paper’s AR(1) prior

We adopt the notation of [Morris, Reich, Thibaud, and Cooley \(2017\)](#) (their Section 3.2). For each spatial knot index $k = 1, \dots, K$ and time index $t = 1, \dots, n_t$, let $\mathbf{w}_{tk}^* \in \mathbb{R}^2$, $z_{tk}^* \in \mathbb{R}$, and $\sigma_{tk}^{2*} \in \mathbb{R}$ denote the standard-normal-scale transformations of the spatial knot, half-normal random effect, and inverse gamma variance scale, respectively. The paper’s AR(1) prior, restated in the notation we will need:

$$\mathbf{w}_{1k}^* \sim N_2(\mathbf{0}, \mathbf{I}_2), \quad z_{1k}^* \sim N(0, 1), \quad \sigma_{1k}^{2*} \sim N(0, 1), \quad (1)$$

$$\mathbf{w}_{tk}^* \mid \mathbf{w}_{t-1,k}^* \sim N_2[\phi_w \mathbf{w}_{t-1,k}^*, (1 - \phi_w^2) \mathbf{I}_2], \quad (2)$$

$$z_{tk}^* \mid z_{t-1,k}^* \sim N[\phi_z z_{t-1,k}^*, 1 - \phi_z^2], \quad (3)$$

$$\sigma_{tk}^{2*} \mid \sigma_{t-1,k}^{2*} \sim N[\phi_\sigma \sigma_{t-1,k}^{2*}, 1 - \phi_\sigma^2], \quad (4)$$

with $|\phi_w|, |\phi_z|, |\phi_\sigma| < 1$.

2.2 The AR(2) extension

Because the three parameter classes share an identical AR structure, we state results for a generic scalar series $\{Y_t\}_{t=1}^{n_t}$ representing any one of $\mathbf{w}_{\cdot k}^*$ (componentwise), $z_{\cdot k}^*$, or $\sigma_{\cdot k}^{2*}$ for a fixed knot k .

Definition 2.1 (Stationary AR(2) prior). A standard AR(2) process satisfies

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \epsilon_t, \quad \epsilon_t \stackrel{iid}{\sim} N(0, \sigma_\epsilon^2), \quad t \geq 3, \quad (5)$$

together with the stationarity conditions

$$|\phi_2| < 1, \quad \phi_1 + \phi_2 < 1, \quad \phi_2 - \phi_1 < 1, \quad (6)$$

and the normalisation $\text{Var}(Y_t) = 1$ for all t (so the skew- t marginal is preserved).

Proposition 2.2 (Induced moments). *Under Proposition 2.1, the Yule–Walker equations together with $\gamma_0 := \text{Var}(Y_t) = 1$ yield*

$$\gamma_0 = 1, \quad \gamma_1 = \frac{\phi_1}{1 - \phi_2}, \quad \gamma_2 = \phi_1 \gamma_1 + \phi_2, \quad \sigma_\epsilon^2 = 1 - \phi_1 \gamma_1 - \phi_2 \gamma_2, \quad (7)$$

all functions of (ϕ_1, ϕ_2) only.

Proof. The autocovariance function $\gamma_h := \text{Cov}(Y_t, Y_{t+h})$ satisfies the Yule–Walker recursion $\gamma_h = \phi_1 \gamma_{h-1} + \phi_2 \gamma_{h-2}$ for $h \geq 1$, with $\gamma_{-h} = \gamma_h$. Setting $h = 1$ and using $\gamma_0 = 1$ gives $\gamma_1 = \phi_1 + \phi_2 \gamma_1$, hence $\gamma_1 = \phi_1 / (1 - \phi_2)$. Setting $h = 2$ gives $\gamma_2 = \phi_1 \gamma_1 + \phi_2 \gamma_0 = \phi_1 \gamma_1 + \phi_2$. Multiplying (5) by Y_t and taking expectations yields $\gamma_0 = \phi_1 \gamma_1 + \phi_2 \gamma_2 + \sigma_\epsilon^2$, so $\sigma_\epsilon^2 = 1 - \phi_1 \gamma_1 - \phi_2 \gamma_2$. $\square$

Proposition 2.3 (Joint factorisation under stationarity). *Assume $\{Y_t\}_{t=1}^{n_t}$ is stationary under (5) with unit marginal variance. Then the joint density factorises as*

$$p(Y_1) = N(0, 1), \quad (8)$$

$$p(Y_2 | Y_1) = N(\gamma_1 Y_1, 1 - \gamma_1^2), \quad (9)$$

$$p(Y_t | Y_{t-1}, Y_{t-2}) = N(\phi_1 Y_{t-1} + \phi_2 Y_{t-2}, \sigma_\epsilon^2), \quad t \geq 3. \quad (10)$$

Proof. Stationarity gives $\text{Var}(Y_1) = 1$, so (8) follows from Gaussianity. For (9): the stationary joint of (Y_1, Y_2) is $N_2(0, \begin{bmatrix} 1 & \gamma_1 \\ \gamma_1 & 1 \end{bmatrix})$, and the conditional of a bivariate Gaussian gives mean $\gamma_1 Y_1$ and variance $1 - \gamma_1^2$. For (10): (5) directly gives $Y_t | (Y_{t-1}, Y_{t-2}) \sim N(\phi_1 Y_{t-1} + \phi_2 Y_{t-2}, \sigma_\epsilon^2)$. $\square$

Remark 2.4. (9) is sometimes replaced in the literature by the AR(1)-style density $N(\phi_1 Y_1, \sigma_\epsilon^2)$, which arises if one assumes (Y_1, Y_2) are independent with $Y_2 \sim N(0, 1)$ and then runs the recursion from $t = 3$. This non-stationary initialisation is consistent with the marginal statements “ $Y_1 \sim N(0, 1)$ and $Y_2 \sim N(0, 1)$ ” read in isolation, but it does *not* preserve $\text{Var}(Y_t) = 1$ for $t \geq 3$ unless σ_ϵ^2 is adjusted away from its Yule–Walker value. The implementation we audit uses (9)’s non-stationary form at $t = 2$ inside the MCMC, while the simulator `simulate_ar2_standard` samples (Y_1, Y_2) jointly from the stationary N_2 . The two are therefore consistent with each other for $t \geq 3$ but disagree on the single density factor at $t = 2$.

3 Audit of the implementation

We label the three deviations identified during the audit as Issue 1 ($t = 2$ conditional), Issue 2 (missing lag-2 term), and Issue 3 (ϕ -proposal asymmetry). Section 3.2–Section 3.3 present the detailed-balance analysis that classifies each.

3.1 Issue 1: the $t = 2$ conditional

The function `get_ar2_conditional_params` in `auxfunctions_ar2.R` previously returned, for $t = 2$, the AR(2) recursion-style form $N(\phi_1 Y_1, \sigma_\epsilon^2)$ rather than the stationary form (9). This was the AR(2) recursion extrapolated backward with a fictitious $Y_0 = 0$, and it disagreed with the joint draw that the simulator `simulate_ar2_standard` performs at $t = 1, 2$. The two coincide only when $\phi_2 = 0$.

Patch. The $t == 2$ branch has been replaced by (9):

```
# Fix 1: stationary AR(2) conditional at t = 2
if (t == 2) {
  g1      <- moments$gamma1
  sd.t2 <- sqrt(moments$gamma0 - g1^2) # = sqrt(1 - gamma1^2)
  return(list(mean = g1 * lag1, sd = sd.t2))
}
```

With this in place, the MCMC density at $t = 2$ agrees with the simulator’s joint draw and the marginal property $\text{Var}(Y_t) = 1$ is preserved by every component of the sampler.

Regression test. `tests/test_fix1_t2_conditional.R` is a closed-form unit test that calls `get_ar2_conditional` at six (ϕ_1, ϕ_2) configurations spanning the interior and the near-boundary regions of the stationarity triangle, and checks the returned (mean, sd) against the Yule–Walker target $(\gamma_1 y_1, \sqrt{1 - \gamma_1^2})$ to machine precision (10^{-10}). On the un-patched code the test fails at five of six configurations (only the $\phi_2 = 0$ row matches); after the patch it passes at all six.

3.2 Issue 2: missing lag-2 term in the Y_t MH ratio

3.2.1 Detailed-balance argument

Under the AR(2) prior, Y_t enters *three* factors of the joint distribution: as the conditioned-on variable at time t , as the lag-1 predictor at time $t+1$, and as the lag-2 predictor at time $t+2$. The full conditional density of Y_t given the rest of the chain and the data is therefore

$$\pi(Y_t | Y_{-t}, \text{data}) \propto L(\text{data} | Y_t) \cdot \underbrace{p(Y_t | Y_{t-1}, Y_{t-2})}_{\text{self}} \cdot \underbrace{p(Y_{t+1} | Y_t, Y_{t-1})}_{\text{lag-1}} \cdot \underbrace{p(Y_{t+2} | Y_{t+1}, Y_t)}_{\text{lag-2}}. \quad (11)$$

For a symmetric Metropolis proposal $Y_t \rightarrow Y_t^c$, detailed balance with target (11) requires the acceptance probability $\alpha = \min(1, R)$ with

$$\log R = \log \frac{L(\text{data} | Y_t^c)}{L(\text{data} | Y_t)} + \sum_{s \in \{t, t+1, t+2\}} [\log p(Y_s | \text{lags}, Y_t = Y_t^c) - \log p(Y_s | \text{lags}, Y_t)]. \quad (12)$$

Proposition 3.1 (Detailed-balance violation in the original code). *The original `update{Tau,Z,Knots}TS_AR2` functions in `update.params.ar2.R` include only the $s = t$ and $s = t+1$ factors in $\log R$. The omitted $s = t+2$ factor enters the joint AR(2) prior whenever $t \leq n_t - 2$. The chain therefore satisfies detailed balance for the modified target*

$$\tilde{\pi}(Y_t | \dots) \propto L(\text{data} | Y_t) p(Y_t | Y_{t-1}, Y_{t-2}) p(Y_{t+1} | Y_t, Y_{t-1}), \quad (13)$$

which is not the AR(2) full conditional.

Corollary 3.2. *The stationary marginal of the chain under (13) does not have lag-2 autocorrelation equal to γ_2 . The empirical lag-2 ACF of a prior-only chain (no data likelihood) under the original code differs from (7) by an $O(1)$ amount.*

3.2.2 Patch

For each of the four call sites (`updateTauTS_AR2` for both `nknots == 1` and `nknots > 1`, `updateZTS_AR2`, and `updateKnotsTS_AR2`) we add a $t+2$ correction parallel to the existing $t+1$ block. The patch for `updateZTS_AR2`:

```
# Fix 2: z.star[k, t] enters the t+2 conditional density as lag-2
if ((t + 2) <= nt) {
  z.star.next2 <- z.star[k, t + 2]
  R <- R +
    ar2_conditional_log_density(
      z.star.next2, t + 2,
      z.star[k, t + 1], can.z.star[k],
      phi1 = phi1, phi2 = phi2, moments = ar2.moments) -
    ar2_conditional_log_density(
      z.star.next2, t + 2,
      z.star[k, t + 1], z.star[k, t],
      phi1 = phi1, phi2 = phi2, moments = ar2.moments)
}
```

For `updateKnotsTS_AR2` the density returns a $K \times 2$ matrix that is aggregated by `sum(...)`, mirroring the existing $t+1$ block. At $t = 1$ the companion $t+1$ block evaluates the $t = 2$ conditional via `get_ar2_conditional_params`, which after the Issue 1 patch returns the stationary form (9). The Fix 1 dependency that this note used to carry inline at the four patch sites has been resolved; the comments now simply point back to Section 3.1.

3.3 Issue 3: ϕ proposal asymmetry — a non-issue

The AR(1) reference `updatePhiTS` samples a ϕ candidate from a *truncated* normal via inverse CDF and includes the Hastings correction

$$\Delta_H = \log P(S; \phi^{(r-1)}) - \log P(S; \phi^{(c)}), \quad P(S; m) = \Phi\left(\frac{b-m}{\tau}\right) - \Phi\left(\frac{a-m}{\tau}\right), \quad (14)$$

for $S = (a, b)$ the AR(1) stationarity interval $(-1, 1)$. The AR(2) implementation `updatePhiAR2TS` instead samples $\phi^{(c)} \sim N(\phi^{(r-1)}, \tau^2)$ from an *untruncated* normal and silently rejects when $\phi^{(c)}$ falls outside the conditional stationarity interval

$$S_1(\phi_2) = (\phi_2 - 1, 1 - \phi_2), \quad S_2(\phi_1) = (|\phi_1| - 1, 1 - |\phi_1|). \quad (15)$$

We initially conjectured that the silent-rejection mechanism is operationally equivalent to truncated sampling and therefore requires the same Hastings correction (14). Proposition 3.3 below shows this conjecture is *wrong*: silent rejection of out-of-region candidates from a symmetric proposal already targets the correct posterior, with no Hastings correction needed.

Proposition 3.3 (Silent-rejection scheme is detailed-balance correct). *Let π be a target density on $\mathbb{R}^d$ supported on a region $S \subset \mathbb{R}^d$ (i.e. $\pi(\phi) = 0$ for $\phi \notin S$). Let $q_0(\cdot | \phi)$ be a symmetric proposal density on $\mathbb{R}^d$. Consider the Metropolis kernel*

$$\Pr(\phi \rightarrow \phi^c) = \begin{cases} q_0(\phi^c | \phi) \min(1, \pi(\phi^c)/\pi(\phi)) & \phi^c \in S, \\ 0 & \phi^c \notin S, \end{cases} \quad (16)$$

with the residual mass $\Pr(\phi \rightarrow \phi)$ chosen so that the total probability sums to one. Then π is invariant for the chain defined by (16).

Proof. We verify detailed balance for $\phi, \phi^c \in S, \phi \neq \phi^c$:

$$\begin{aligned} \pi(\phi) q_0(\phi^c | \phi) \min(1, \pi(\phi^c)/\pi(\phi)) &= q_0(\phi^c | \phi) \min(\pi(\phi), \pi(\phi^c)) \\ &= q_0(\phi | \phi^c) \min(\pi(\phi^c), \pi(\phi)) \\ &= \pi(\phi^c) q_0(\phi | \phi^c) \min(1, \pi(\phi)/\pi(\phi^c)), \end{aligned}$$

where the middle equality uses the symmetry $q_0(\phi^c | \phi) = q_0(\phi | \phi^c)$. For $\phi^c \notin S$ the move has probability zero on both sides because $\pi(\phi^c) = 0$. Detailed balance holds pointwise on $S \times S$, hence π is invariant. $\square$

Proposition 3.4 (The proposed Hastings “fix” shifts the target). *If the kernel (16) is modified by including the Hastings factor $P(S; \phi)/P(S; \phi^c)$ in the acceptance ratio while retaining silent rejection, the resulting chain is invariant not for π but for*

$$\tilde{\pi}(\phi) \propto \frac{\pi(\phi)}{P(S; \phi)}. \quad (17)$$

Because $P(S; \phi)$ is maximised at the interior of S and decreases toward its boundary, $\tilde{\pi}$ shifts mass toward the boundary relative to π .

Proof. With acceptance ratio $\alpha^\dagger(\phi, \phi^c) = \min(1, [\pi(\phi^c)/\pi(\phi)] \cdot [P(S; \phi)/P(S; \phi^c)])$, detailed balance for the candidate target $\tilde{\pi}$ requires

$$\tilde{\pi}(\phi) q_0(\phi^c | \phi) \alpha^\dagger(\phi, \phi^c) = \tilde{\pi}(\phi^c) q_0(\phi | \phi^c) \alpha^\dagger(\phi^c, \phi),$$

which after the substitution $\tilde{\pi}(\phi) = \pi(\phi)/P(S; \phi)$ collapses to

$$q_0(\phi^c \mid \phi) \min(\tilde{\pi}(\phi), \tilde{\pi}(\phi^c)) = q_0(\phi \mid \phi^c) \min(\tilde{\pi}(\phi^c), \tilde{\pi}(\phi)),$$

which holds by symmetry of q_0 . To see the direction of the shift, write $P(S; \phi) = \Phi(\frac{b-\phi}{\tau}) - \Phi(\frac{a-\phi}{\tau})$ for $S = (a, b)$. By symmetry of Φ , $P(S; \phi)$ is maximised at $\phi = (a + b)/2$ and strictly decreasing as $|\phi - (a + b)/2|$ increases. Hence $1/P(S; \phi)$ is minimised at the centre of S and increases toward the boundary, shifting mass of $\tilde{\pi}$ away from the centre relative to π . $\square$

Corollary 3.5. *The original `updatePhiAR2TS` (silent rejection, no Hastings term) is detailed-balance-correct for the posterior of (ϕ_1, ϕ_2) restricted to the $AR(2)$ stationarity region. The Hastings correction was removed and the production code now uses the silent-rejection scheme. An optional follow-up, deferred unless mixing near the boundary becomes a concern, is to refactor the function to mirror `updatePhiTS`’s truncated-sampling scheme; in that variant the Hastings correction is mandatory.*

3.4 Why the initial Fix 3 test was misleading

The paired test `tests/test_fix3_phi_hastings.R` placed the truth at $(\phi_1^*, \phi_2^*) = (0.45, 0.50)$, approximately 90% of the way to both edges of S_1 and S_2 . Near the boundary, $\tilde{\pi}$ in (17) concentrates near the truth more tightly than the true posterior π , so the “patched” chain produced posterior medians closer to the truth than the original chain *by coincidence*. The reported bias-reduction ratio of 0.48 on ϕ_1 across 12 replicates is well within Monte-Carlo noise: per-chain posterior medians vary by ~ 0.10 , giving a standard error on the mean of ~ 0.03 – larger than the signal claimed.

A diagnostic that would have caught the issue is the same test with the truth placed at $(\phi_1^*, \phi_2^*) = (0, 0)$, away from any boundary. Under Proposition 3.4 the “patched” chain should sit *farther* from zero than the original chain in this regime; the unpatched chain is correct. We did not run this diagnostic before applying the patch.

4 Patches and regression-test infrastructure

4.1 Patches applied

- **Issue 1 ($t = 2$ conditional):** *applied* (one-line change in `get_ar2_conditional_params`). The $t = 2$ branch now returns the stationary form $N(\gamma_1 y_1, 1 - \gamma_1^2)$.
- **Issue 2 (lag-2 MH term):** applied at four sites in `code/R/ar2/update_params_ar2.R`.
- **Issue 3 (ϕ Hastings correction):** *reverted*. The Hastings adjustment was removed; `updatePhiAR2TS` is back to the silent-rejection scheme, which by Proposition 3.3 targets the correct posterior.

4.2 Regression tests

Five tests live in `code/R/ar2/tests/`, each runnable via `Rscript test_*.R` and returning exit code 0/1.

Test file	Targets	Layer / what it catches
<code>test_fix1_t2_conditional.R</code>	Issue 1	unit: $(\gamma_1 y_1, \sqrt{1 - \gamma_1^2})$ to machine precision
<code>test_fix2_lag2_mh.R</code>	Issue 2	math: ACF of prior-only chain vs. Yule–Walker
<code>test_fix2_production.R</code>	Issue 2	integration: real <code>updateZTS_AR2</code> with zero data
<code>test_fix2_grep.R</code>	Issue 2	structural: four <code>(t + 2) <= nt</code> blocks present
<code>test_fix3_phi_hastings.R</code>	Issue 3 (revert)	structural + sanity (no <code>pnorm</code> in <code>updatePhiAR2TS</code>)

The Issue 1 test is a closed-form unit check against the Yule–Walker target at six (ϕ_1, ϕ_2) configurations. The Issue 2 integration test was verified to fail (lag-2 ACF error jumps from 0.014 to 0.284) when the patch was temporarily commented out, then pass once restored. The Issue 3 test was rewritten in the same round as the revert: it now parses the body of `updatePhiAR2TS` and asserts that no `pnorm(.)` calls appear – the signature of the wrongly-shipped Hastings adjustment – together with a sanity assertion that the silent-rejection chain recovers an interior truth.

4.3 Test results, current state

Issue 2 math test	buggy chain	patched chain
target lag-1 ACF $\gamma_1 = 0.500$		
target lag-2 ACF $\gamma_2 = -0.050$		
empirical lag-1 ACF	+0.608	+0.498
empirical lag-2 ACF	+0.222	−0.054
empirical marginal variance	1.142	0.998

The buggy chain inflates the marginal variance by 14% and gets the lag-2 autocorrelation *wrong in sign*; the patched chain matches the Yule–Walker targets to four decimal places.

Issue 2 integration test	lag-1 ACF error	lag-2 ACF error
patched (current code)	0.0002	0.0145
reverted (smoke check)	0.1121	0.2840

5 Current state and next steps

The end-state of the audit is:

1. **Issue 1 is closed.** The $t = 2$ branch of `get_ar2_conditional_params` now returns the stationary form $N(\gamma_1 y_1, 1 - \gamma_1^2)$. The simulator and the MCMC density now agree on the joint of (Y_1, Y_2) , and $\text{Var}(Y_t) = 1$ holds across every component of the sampler. The unit test `test_fix1_t2_conditional.R` verifies the returned (mean, sd) against the Yule–Walker target to machine precision at six (ϕ_1, ϕ_2) configurations.
2. **Issue 2 is closed.** The lag-2 MH correction is applied at all four sites, the math test verifies the AR(2) stationary marginals to four decimal places, the integration test catches the regression on the actual production function, and the structural grep test guards the four sites collectively against bulk deletion. Proposition 3.1 establishes that the correction is mathematically required for the AR(2) full conditional.

3. **Issue 3 is closed by reverting.** The Hastings adjustment has been removed; `updatePhiAR2TS` is back to the silent-rejection scheme established as correct in Proposition 3.3. The replacement test (`test_fix3_phi_hastings.R`) parses the body of `updatePhiAR2TS` and asserts that no `pnorm(.)` calls remain — the structural signature of the previously shipped Hastings adjustment — and additionally runs a sanity chain that recovers an interior truth.

Optional follow-up. If mixing near the AR(2) stationarity boundary becomes a concern in practice, refactor `updatePhiAR2TS` to mirror `updatePhiTS`: sample $\phi^{(c)}$ from a truncated normal via inverse CDF and include the Hastings correction (14). The silent-rejection scheme and the truncated-sampling scheme have the same stationary distribution; the latter avoids wasted draws on out-of-region candidates and typically mixes faster near boundaries, at the cost of a slightly more involved update step.

6 Conclusion

A careful detailed-balance audit of the AR(2) extension to the Morris, Reich, Thibaud, and Cooley (2017) space-time skew- t model produced two confirmed bugs and one rejected “bug”. The first confirmed bug — the $t = 2$ conditional density used inside the MCMC — broke the simulator-versus-MCMC agreement on the joint of (Y_1, Y_2) and was patched by replacing the recursion-style $N(\phi_1 y_1, \sigma_\epsilon^2)$ with the stationary form $N(\gamma_1 y_1, 1 - \gamma_1^2)$. The second — the missing lag-2 term in the Y_t Metropolis–Hastings ratio — was patched at all four call sites in `update_params_ar2.R`. The rejected “bug” was the apparent asymmetry of the ϕ proposal: the silent-rejection scheme in the original `updatePhiAR2TS` was correct all along, the Hastings adjustment we initially applied targets a boundary-shifted posterior, and that adjustment has been reverted.

The lesson for the broader project is methodological: empirical “bias-reduction” on a single configuration is not a sufficient guard against incorrect MCMC patches. Detailed balance must be checked from first principles for any change to a Metropolis–Hastings acceptance ratio, and regression tests must include both near-boundary and far-from-boundary truths to expose target-shifting errors that flatter themselves in one regime and fail in the other.

References

Samuel A. Morris, Brian J. Reich, Emeric Thibaud, and Daniel Cooley. A Space-Time Skew- t Model for Threshold Exceedances. *Biometrics*, 73(3):749–758, September 2017. ISSN 0006-341X. doi: 10.1111/biom.12644. URL <https://doi.org/10.1111/biom.12644>.